

Cognitive Accessibility in Generative AI Interfaces: A Systematic Review

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ABSTRACT

Generative Artificial Intelligence (GAI) represents a significant shift in the way people interact with technology, with a growing impact in sectors such as education, communication, and content creation. Although its applications promise relevant benefits for people with cognitive disabilities, such as language simplification or support for task organization, there are also significant accessibility challenges that remain unexplored. This paper presents the results of a systematic literature review focused on the cognitive accessibility of textual GAI interfaces. The main objective was to identify the current cognitive accessibility challenges inherent in textual GAI interfaces and to synthesize the solutions proposed to promote a more inclusive interaction. To this end, seven studies selected between 2020 and 2024 from four academic databases (Scopus, Web of Science, IEEE Xplore, and ACM Digital Library) were analyzed. The analysis reveals that current interfaces impose excessive cognitive load and suffer from systemic lack of predictability and user scaffolding due to their reliance on complex prompt engineering and low explainability. Emerging proposals based on inclusive design, personalization, multimodal interfaces, and ethical frameworks are also identified. The paper concludes by highlighting the need for more empirical studies, specific design guidelines, and validated evaluation methods to ensure that generative technologies are truly accessible to all.

KEYWORDS

Human-Computer Interaction (HCI), Accessible Interfaces, Generative Artificial Intelligence (GAI), Cognitive Disability, Cognitive Accessibility, Inclusive Design, Accessibility Assessment

1. Introduction and Related Work

Generative Artificial Intelligence (GAI) is defined by Lim et al. (2023) as “*a technology that (i) leverages deep learning models to (ii) generate human-like content (e.g., images, words) in response to (iii) complex and varied prompts (e.g., languages, instructions, questions)*”. In recent years, GAIs have significantly influenced many areas of daily life, transforming how we search for information, write, program, and even perform research.

GAI tools hold significant promise for people with cognitive disabilities. For students with communication disabilities, such as dyslexia or cerebral palsy, GAI could facilitate written communication by offering spelling and grammar correction, sentence rephrasing, and even full-text generation (de Haas, 2023; Heidt, 2024; Bradshaw, 2023). It can also help neurodiverse students understand complex texts by summarizing, simplifying, and clarifying textual content (Heidt, 2024; Leos, 2023). In addition, people with Attention Deficit Hyperactivity Disorder (ADHD) may benefit from customizable reminders, motivational prompts, and support in organizing daily routines (Leos, 2023).

The emergence of GAI interfaces represents a fundamental paradigm shift in Human-Computer Interaction (HCI), moving away from structured, menu-based systems toward open-ended, prompt-based interactions (Nielsen, 2023). While the inclusive potential of GAI is frequently highlighted, this new interaction model introduces significant cognitive challenges that remain critically underexplored. Traditional interfaces provided predictability through visible system status and predefined options, thereby managing cognitive load. In contrast, GAI requires users to formulate abstract queries, interpret non-deterministic responses, and iterate through trial and error (Tankelevitch et al., 2023). Instructions might be confusing, visual interfaces look somehow chaotic with visual noise, the way the information is presented imposes a significant extrinsic load on people with cognitive disabilities and it is necessary to reduce it. The dependence on complex prompt formulation reveals a systemic lack of scaffolding mechanisms and inherently increases overall cognitive load, severely undermining predictability. Predictability is key to reducing extrinsic cognitive load, because when the user knows what to expect, they do not need to invest working memory or mental effort in figuring out how the system works (Daronnat et al., 2021). With appropriate scaffolding, the task can be made more manageable by dividing it into steps and reducing its complexity (intrinsic load); it can remove noise by reducing unnecessary decisions and clarifying the task (extrinsic load); and it can increase understanding and guided interaction (germane load) (Hollender et al., 2010).

Despite these widespread structural problems, current literature has largely failed to evaluate GAI interfaces using an HCI framework. To date, the few studies that assess cognitive accessibility often focus only on isolated issues such as unreliable speech-to-text input (Guo et al., 2020) or high output reading levels (Roegiest and Pinkosova, 2024). GAI systems must be evaluated using HCI frameworks based on the basic principles of cognitive accessibility before it can be determined whether the potential identified in these applications for people with cognitive disabilities is close to becoming a reality. Given that the new GAI paradigm systematically challenges core principles of cognitive accessibility, the structural exclusion of users with disabilities from these emerging technologies is a significant and pressing concern. Accessibility is a fundamental prerequisite for equitable participation in digital environments, and that exclusion from emerging technologies such as GAI may exacerbate existing social and educational inequalities. As Morris (2020) emphasizes, when digital systems are not designed with disability in mind, they risk excluding entire communities from access to essential services. In that sense, Alshaigy and Grande (2024) critically examine the persistent inequities between the rapid development of GAI and the accessibility needs of users, arguing these tools are predominantly developed by able-bodied designers alongside the absence of design guidelines specifically tailored for GAI accessibility. Addressing these persistent conceptual and structural becomes a pressing issue for ensuring more accessible AI.

Some recent works have begun to examine this gap from an inclusive research perspective. In scoping review of the state of the art, Hemsley et al. (2024) report a positive trend in the adoption of innovative methodologies such as GAI to elicit the perspectives

of individuals with communication disabilities. These approaches contribute to a deeper understanding of the risks users face and the strategies they employ when navigating social networks. Similarly, Rice and Dunn (2023) explore the use of AI to support young people with disabilities, indicating that while this growth in use presents opportunities for empowerment, it is important that researchers remain critical of these technologies and actively advocate for inclusive practices.

To the best of our knowledge, there are no systematic literature reviews that address the inclusive design of GAI interfaces with a specific focus on cognitive accessibility. This article aims to address that gap through a systematic literature review (SLR) that synthesizes current knowledge at the intersection of GAI interfaces and cognitive accessibility. This synthesis is conducted through a conceptual framework that links design elements to the core principles of cognitive accessibility. The key objectives of this study are to: (1) analytically evaluate the current interface design characteristics and the main challenges users with cognitive disabilities face when interacting with GAI, and (2) analyze the solutions proposed to make GAI more accessible.

To achieve this, the remainder of the article is structured as follows. Section 2 presents the methodology used for this SLR. Section 3 describes the results, organized according to the proposed research questions. Section 4 discusses the implications of our findings, including strengths, limitations, and directions for future research. Finally, Section 5 offers concluding remarks.

2. Methodology

To ensure a rigorous and transparent review process, this study followed a systematic methodology based on the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement (Moher et al., 2009), which provides a structured approach to help researchers report their reviews in a clear, transparent, and replicable manner (Creswell and Clark, 2017). The aim was to identify and analyze scientific literature that addresses the cognitive accessibility of textual Generative AI (GAI) interfaces. This systematic literature review (SLR) addresses a set of clearly defined research questions by analyzing peer-reviewed publications from 2020 to 2024, retrieved from four major academic databases in the field of Computer Science: ACM, IEEE, Web of Science (WoS), and Scopus.

Following the PRISMA framework, we organized the review process into four sequential phases:

- **Identification:** An automatic search was conducted using a query string designed to capture the key concepts related to the research questions across the selected databases.
- **Screening:** The initial set of retrieved studies was filtered by reviewing titles, abstracts, and keywords to identify those that aligned most closely with the objectives of the review.
- **Eligibility:** Full-text versions of the shortlisted articles were assessed by authors to extract relevant categories and descriptors for in-depth analysis.
- **Inclusion:** The final set of studies was selected based on a quality assessment and metadata analysis of the identified descriptors.

This section is structured as follows. Section 2.1 presents the research questions that guided the review. Section 2.2 presents the eligibility criteria, followed by the description of data sources and search strategy (Section 2.3), and study selection process (Section

2.4). Finally, Section 2.5 explain the synthesis procedures used to analyze and interpret the results.

2.1. Research questions

Our systematic literature review was guided by the following research questions:

- RQ1: What are the current design characteristics and the main cognitive or intellectual accessibility challenges (or barriers) identified in the literature regarding textual Generative AI interfaces?
- RQ2: What solutions have been proposed to improve the accessibility of textual generative AI interfaces for individuals with cognitive disabilities?

These questions aim to structure the review across two essential dimensions. RQ1 is essential for understanding how generative AI interfaces are currently designed, identifying both the predominant design patterns and the key barriers that users with cognitive disabilities face when interacting with these technologies. RQ2 is critical for mapping the proposed strategies and interventions aimed at enhancing accessibility in generative AI interfaces, contributing to more inclusive and user-centered design approaches.

2.2. Eligibility Criteria

Study selection was guided by the PICOC (Population, Intervention, Comparison, Outcomes, Context) framework (Carrera-Rivera et al., 2022), which is appropriate for structuring systematic reviews in the field of human-computer interaction and accessibility. This framework was applied as follows:

- Population: Generative AI tools that have a user interface and generate textual content. This criterion allows us to focus on systems that involve direct interaction between the user and the AI model, which is essential for evaluating interface accessibility.
- Intervention: Studies that explore or evaluate aspects related to the cognitive accessibility of these interfaces. Priority is given to studies that specifically address barriers, supports, or design proposals aimed at people with cognitive difficulties.
- Comparison: Tools that do not incorporate cognitive accessibility criteria, used as a contrast or reference point when discussing design limitations or gaps.
- Outcome: Identification of barriers, best practices, recommendations, or findings related to cognitive accessibility. This criterion seeks to ensure that the selected studies contribute relevant and applicable knowledge to this specific field.
- Context: People with cognitive or intellectual disabilities who could benefit from the use of more accessible generative AI tools. This criterion reflects the overall objective of the study: to promote the inclusion of users with diverse cognitive needs.

To ensure consistency and academic quality, the following inclusion and exclusion criteria were applied:

- Time range (2020–2024): This period includes the years following the popularization of deep learning-based generative AI models, such as GPT-3 and its successors, which have led to the design of new interfaces focused on natural language. The systematic review began in early 2025; therefore, only studies that were pub-

licly available by the end of 2024 were eligible for inclusion. This includes peer-reviewed journal and conference papers officially published during that time, as well as *preprints* and “online first” articles made accessible in 2024, even if their final publication date appears as 2025. This approach ensures a consistent and transparent time frame while allowing for the inclusion of the most recent and relevant research.

- Language (English): Selection was restricted to studies published in English, the predominant language in the international literature on AI and accessibility.
- Study type: Empirical studies (such as case studies and usability tests), theoretical reviews, and articles proposing or evaluating specific interface designs, were included. This methodological diversity ensures both conceptual depth and practical insights.
- Publication type: Only peer-reviewed journal articles and conference papers were considered, to guarantee academic rigor and reliability.
- Content: Studies had to focus specifically on textual generative AI interfaces for cognitive disability, as the focus of this review is on natural language interactions. Articles focusing exclusively on other disability types (e.g., visual or motor impairments), or on generative systems that do not involve user interfaces (e.g., backend-only applications), were excluded.
- Minimum length (6 pages): To ensure sufficient methodological and argumentative development, only articles of at least six pages were included. Short communications, posters, and extended abstracts were excluded, as they typically do not contain sufficient methodological information or detailed results.

These criteria were defined to ensure the relevance, consistency, and quality of the included studies, in line with the research objectives focused on the cognitive accessibility of generative AI textual interfaces.

2.3. Data Sources and Search Strategy

To ensure broad, rigorous, and representative coverage of the scientific literature in the field of generative artificial intelligence and cognitive accessibility, four leading academic databases were selected:

- Scopus: One of the most comprehensive multidisciplinary databases, with extensive coverage of publications in science, technology, engineering, and social sciences.
- IEEE Xplore: Provides access to high-quality technical literature in engineering, computing, and computer science, including publications from key conferences on accessibility and HCI.
- ACM Digital Library: A specialized database for computer science and information technology, with a strong focus on HCI, interaction design and user-centered technologies.
- Web of Science (WoS): Offers a consolidated platform for searching scientific literature across various disciplines, including educational technology, cognitive sciences, and digital accessibility.

These databases were chosen for their relevance to the fields of computer science, HCI, digital accessibility, and emerging technologies. All include peer-reviewed publications and provide a high level of indexing of relevant journals and conference proceedings. The combined use of these four sources allowed to identify relevant studies from diverse

disciplinary perspectives, reducing the risk of bias that may arise from relying on a limited number of databases.

To ensure the retrieval of relevant studies aligned with our research questions, we defined a search string composed of three main sets of terms:

- (1) Terms related to generative technologies, such as “*Generative AI*” and “*Generative Artificial Intelligence*”.
- (2) Terms referring to user experience and interface-related aspects, including “*User experience*”, “*Interface*”, “*Human Computer Interaction*”, as well as wildcard terms such as “*Usab**” and “*Accessib**” to capture variations like usability/usable and accessibility/accessible.
- (3) Terms associated with cognitive and neurodevelopmental conditions, such as “*impairment**”, “*Autis**”, “*Neurodivers**”, and “*disab**”.

The three groups of keywords were combined using the Boolean AND operator, ensuring that only studies addressing all conceptual areas were retrieved. Specifically, a publication had to include at least one term from each group to be considered in the initial search results. The structure of the query followed the pattern: (1) AND (2) AND (3).

The final search string used for automatic querying across the four selected digital libraries (ACM, IEEE, Web of Science, and Scopus) was constructed as follows:

```
("Generative AI" OR "Generative Artificial Intelligence") AND ("User  
experience" OR "Interface" OR "Human Computer Interaction" OR "Usab*"  
OR "Accessib*") AND ("impairment*" OR "Autis*" OR "Neurodivers*" OR  
"disab*")
```

This strategy was designed to capture literature at the intersection of generative AI technologies, interface usability and accessibility, and cognitive or intellectual disabilities. The process of refining the query was rigorous and iterative, aimed at balancing the inclusion of relevant synonyms and keywords frequently used in the field, while accounting for the limitations of the search text fields in each database (as not all databases support the use of wildcards such as the asterisk).

2.4. Study Selection Process

The study selection process followed the guidelines of the PRISMA framework, with the aim of ensuring a rigorous, transparent, and replicable methodology. To efficiently manage bibliographic references and organize the different phases of study screening, we used Rayyan (Ouzzani et al., 2016), a free web-based tool designed specifically for systematic and narrative reviews. Rayyan supports semi-automated abstract and title screening, offers an intuitive interface, and facilitates collaboration among multiple reviewers by enabling task distribution, custom labeling, and transparent tracking of inclusion decisions.

Citation files retrieved from each database were imported into Rayyan which extracted different metadata such as title, authors, publication year, and automatically computed attributes such as article language. These data populated filterable facets in the Rayyan interface, aiding the organization and filtering of results. We also established a set of agreed hashtags (e.g., #no-cognitive-disability, #no-user-interface) to consistently flag reasons for inclusion or exclusion, thus contributing to the transparency and reliability of the review process.

The PRISMA process was conducted in four main phases:

- Phase 1: Identification. The initial search in the four selected academic databases (Scopus, Web of Science, IEEE Xplore, and ACM Digital Library) provided a total of 349 records. Each database was searched by one author to ensure traceability.
- Phase 2: Elimination of duplicates: After merging all retrieved records, 35 duplicates were automatically detected and removed using Rayyan’s built-in tools, resulting in 314 unique articles for screening.
- Phase 3: Title and abstract screening: All 314 articles were independently reviewed by two authors based on their title and abstract. Each article was categorized as “Include,” “Exclude,” or “Doubt,” with justifications documented for the latter two categories. Reviewers worked in parallel, and discrepancies were discussed in joint meetings. When consensus was not reached, a third reviewer was consulted. As a result, 295 articles were excluded, and 19 articles were retained for full-text screening. The high exclusion rate can be attributed to the widespread mention of Generative AI (GAI) in contemporary research, often without a direct focus on user-facing interfaces or accessibility. Common reasons for exclusion included:
 - Studies that mentioned GAI without analyzing its interface design.
 - Studies on cognitive accessibility that did not specifically address GAI with a textual interface.
 - Works with no direct user interaction with GAI tools.
 - Research focused on visual or motor accessibility, rather than cognitive or neurodevelopmental aspects.
 - Articles centered on GAI for image or code generation, rather than text.
 - Articles discussing the use of GAI to assist in interface design, but not evaluating the accessibility of generative tools themselves.
- Phase 4: Full-text screening: The 19 remaining articles were analyzed to verify that they met the eligibility criteria described in Section 2.2 and to extract relevant information for analysis. The same approach than the previous phase was used, with two authors fully reviewing half of the articles with particular care on the eligibility criteria as stated (i.e, the clarity of the objectives, methodological rigor, and relevance of the results for cognitive accessibility). During this phase, 12 articles were eliminated, resulting in a total of 7 articles to be included in the final review.

The entire selection process is visually summarized in Figure 1, which presents the flowchart with the number of articles retained and excluded at each phase.

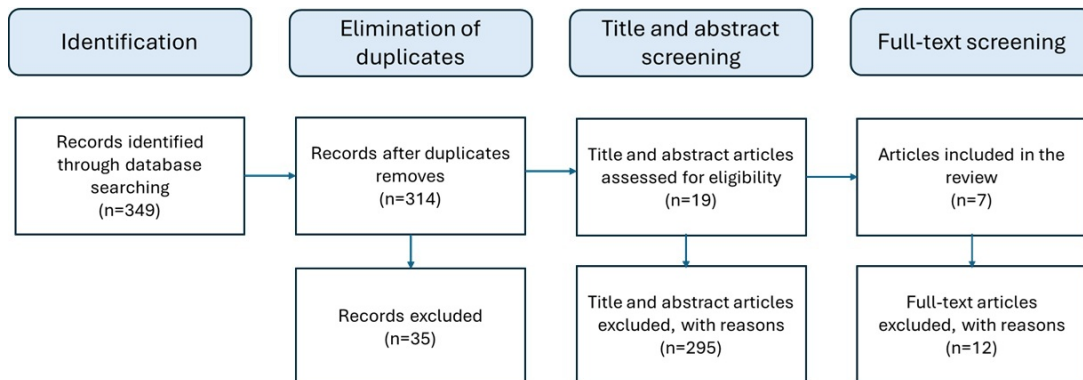


Figure 1. Flow diagram of the study selection process

2.5. Data Synthesis

Data synthesis was carried out collaboratively by the authors during phase 4 of the selection process and discussed collectively during the full-text screening phase. A shared spreadsheet was used to systematically extract and cross-check data among reviewers, based on the following categories:

- Basic bibliographic information: including authors, date, title, type of article, pages, and additional editorial details as applicable.
- Study objectives: a summary of the research aims and primary contributions of each study.
- Methodology: description of the study design, such as whether it involved empirical research (qualitative or quantitative), conceptual frameworks, case studies, or position papers.
- Key findings: synthesis of the main results related to GAI interfaces and their implications for cognitive accessibility, including how each research question (RQ) was addressed.
- Cognitive accessibility considerations: specific ways in which cognitive accessibility was incorporated, evaluated, or discussed in the study. This data extraction included an analytical mapping of findings to HCI principles (cognitive load, predictability, and scaffolding) ensuring we moved from descriptive synthesis to analytical interpretation by evaluating how design elements supported or violated these foundational principles.
- Study limitations: aspects not addressed by the research, limitations in scope or methodology, or areas identified by the authors as needing further work.

These categories were defined to ensure a comprehensive comparison across studies and to directly align the findings with our research questions. For each paper, the extracted data also included a mapping to the relevant RQ(s) to enable focused analysis. Any discrepancies in the data extraction process were resolved through discussion among the authors.

The extracted data informed the synthesis presented in Table 1 and contributed to the discussion in Section 4. Where applicable, studies were also grouped to identify patterns, gaps, and challenges across the literature.

3. Results

This systematic literature review (SLR) identified and analyzed seven peer-reviewed studies selected from conferences and academic journals in the fields of Human-Computer Interaction (HCI), digital accessibility, and educational technologies. The included studies showcase a diversity of methodological approaches, ranging from conceptual proposals and web accessibility audits to case studies, qualitative interviews, and prototype evaluations. Table 1 provides a concise summary of the included studies, detailing their main goals, methodologies, and limitations. Below is a more detailed overview of each work:

- Duarte et al. (2024) explore the grand challenge of the implications of Artificial Intelligence (AI) in the field of HCI, highlighting three key areas: interfaces and interaction paradigms; ethical approaches and design; and the promotion of equity, diversity, and inclusion. The authors propose a long-term research agenda for Brazil (covering the period 2025 to 2035) that includes short, medium, and long-

Author(s)	Year	Title	Type	Main goal	Methodology	Main limitations
Duarte et al.	2024	<i>GrandIHC-BR 2025-2035-GC6: Implications of AI in HCI: A Discussion on Paradigms Ethics and Diversity Equity and Inclusion</i>	Conference (Brazilian Symposium on Human Factors in Computing Systems)	Propose a research agenda on inclusive and ethical HCI in the AI era	Theoretical/conceptual	No empirical data; limited generalization beyond Brazilian context
Liu et al.	2024	<i>Understanding students' perspectives, practices, and challenges of designing with AI in special schools</i>	Conference (International Symposium of Chinese CHI)	Explore how students with special needs interact with GAI tools	Case study with workshops and interviews	Small sample; short exposure to tools; limited transferability
Roomkham & Sitbon	2024	<i>Restarting the conversation about conversational search: exploring new possibilities for multimodal and collaborative systems with people with intellectual disability</i>	Conference (Conference on Human Information Interaction and Retrieval)	Investigate multimodal search interfaces for people with intellectual disabilities	Ethnographic study (WoZ + interviews)	Prototype not scalable; limited number of sessions; human-operated system limits insights into interaction with fully automated systems
Haroony & Dogar	2024	<i>TwIPS: A large language model powered texting application to simplify conversational nuances for autistic users</i>	Conference (ACM SIGACCESS Conference on Computers and Accessibility)	Design and evaluate a messaging app powered by LLMs for autistic users	Lab study with simulated interaction	Controlled setting; university sample only; limited ecological validity
Acosta-Vargas et al.	2024	<i>Addressing Web Accessibility Challenges with Generative Artificial Intelligence Tools for Inclusive Education</i>	Conference (International Conference on eDemocracy & eGovernment)	Evaluate accessibility of 20 GAI homepages using WCAG 2.2	Automated tool-based audit (WAVE)	Focus only on homepages; relies on automatic tools
Acosta-Vargas et al.	2024	<i>Generative Artificial Intelligence and Web Accessibility: Towards an Inclusive and Sustainable Future</i>	Journal (Emerging Science Journal)	Broader audit of 50 GAI tools; explores compliance and policy implications	Combined auto/manual audit	Limited depth in cognitive-specific issues; emphasis on visual barriers
Pierrès et al.	2025 - Online first in 2024	<i>Exploring the role of generative AI in higher education: Semi-structured interviews with students with disabilities</i>	Journal (Education and Information Technologies)	Understand how students with disabilities use ChatGPT in education	Semi-structured interviews	Mostly neurodiverse sample; no longitudinal data; limited generalization

Table 1. Summary of included studies, their goals, methodologies, and limitations

- term actions aimed at promoting socially conscious and critical AI development. Therefore the study is conceptual and it lacks empirical evaluation or user testing. However it identifies important concerns related to cognitive accessibility, such as the difficulty of keyword-based searching for users with cognitive impairments, which demands abstract reasoning and search strategy skills. Other relevant insights include the risk of information overload among users with memory difficulties, and the observation that systems like ChatGPT may provide inaccurate information, requiring users to exercise discretion and critical thinking.
- Liu et al. (2024) present an Artificial Intelligence Generated Content (AIGC) learning experience of seven students with disabilities from two special schools who participated in a series of game design workshops. They developed and tested the insights from a text-to-image GAI workshop, conducting a user study through observations and semi-structured interviews to assess students' knowledge of AI and digital literacy. Data was collected through video analysis, questionnaires, the evaluation of the design outputs and informal interviews. Findings showed that participants benefited from multimodal interaction combining visuals, speech, and text to clarify intent and support understanding: using images first, followed by speech and text, improved comprehension and stimulated engagement. Additionally, the collaborative workshop setting supported awareness of peers' activities and the discovery of personal interests. Limitations included the small participant pool due to time and resource constraints, a short workshop duration, and the exclusive focus on students' perspectives. The authors suggest involving teachers and parents in future studies to broaden insights.
 - Roomkham and Sitbon (2024) investigate multimodal and collaborative search systems designed for people with intellectual disabilities. Authors conducted an ethnographic study with 20 participants with intellectual disability using an ability-centered approach and a Wizard-of-Oz (WoZ) prototype of a multimodal conversational search system (MMCS), which allowed interactions through speech, text, and images. Data collection included audio/video recordings of semi-structured interviews, surveys, and qualitative thematic analysis. Results revealed that individuals with intellectual disability often prefer visual information, such as images and videos, and they are also often more comfortable expressing themselves using pictures or gestures or with the support of others. While mainstream speech recognition is advanced, participants experienced anxiety when interacting with systems that lacked support for stammering or required sustained patience. The exploratory nature of the study and its relatively small participant pool, coupled with the use of a human-operated system, limited insights into interactions with fully automated systems. Future work is needed to address system scalability and response times in real-world contexts.
 - Haroon and Dogar (2024) design and evaluate TwIPS, a LLM-driven text messaging application intended to help autistic users interpret and convey emotional tone and nuanced meaning in text-based communication. A user study was conducted with eight autistic participants in a lab setting, using scripted conversational scenarios and simulated interaction with an AI persona. Data collection included semi-structured interviews, surveys, and thematic analysis. Findings indicate that TwIPS supported users by: a) helping them interpret tone and meaning in incoming messages; b) aligning the emotional tone of outgoing messages with intended meaning; and c) generating alternate phrasings to avoid misinterpretation. However, the participant sample was limited to university students, reducing age and background diversity. Additionally, interactions occurred in a controlled

lab environment with simulated conversations, which may not fully reflect real-world texting dynamics. Further research is recommended to explore deployment in naturalistic settings.

- Acosta-Vargas et al. (2024a) conduct a preliminary accessibility evaluation of 20 widely used GAI applications in the educational domain, including tools like ChatGPT, Gemini, Perplexity, and DALL-E. The study used the WAVE automated tool to analyze the accessibility of each application’s homepage. Findings revealed that 79% of the evaluated tools do not comply with the perceptible principle, which is related to problems in aspects such as easy readability of the font, text alternatives, information and relationships, sensory characteristics, and minimum contrast. The operable principle represents 13% of the total errors, including problems with keyboard accessibility, skip links, and labeling. The understandable principle constitutes 5% of found problems, mainly regarding language definitions and instruction clarity. Finally, the robust principle covers the final 3% of errors, due to limited support for screen readers. The evaluation focused only on the home pages and relied exclusively on automated tools, which may overlook critical nuances. The authors emphasize the need for combined manual and automated evaluations to achieve a more complete accessibility assessment.
- Acosta-Vargas et al. (2024b) expand on their previous work by evaluating the accessibility of 50 GAI applications, using this time WCAG 2.2 success criteria and a variation of the Website Accessibility Conformance Evaluation Methodology (WCAG-EM) 1.0 (W3C Working Group, 2014). The evaluation included both automated testing (WAVE) and manual inspection. Results identified significant variability across tools, with many showing contrast errors, missing structural elements, inconsistent use of ARIA roles, and low compliance with accessibility principles. The authors note that GAI content is often written at a college reading level, which may reduce accessibility. They also raise concerns about algorithmic bias and lack of transparency.
- Pierrès et al. (2025) explore the way students with disabilities use and perceive ChatGPT as a tool in higher education. Semi-structured interviews with 33 students (including those with neurodivergence, visual impairments, chronic illnesses, hearing impairments, and mental health conditions) revealed that ChatGPT provides valuable assistance in writing, reading, studying, and organizing tasks. Writing was identified as a major challenge, especially for students with dyslexia, dysorthographia, or ADHD, who reported difficulties with spelling, grammar, vocabulary, and initiating the writing process. While the study offers rich insights, it has limitations: the sample may be biased toward students who had already requested accommodations and had prior experience with ChatGPT. Additionally, the requirement that participants had used ChatGPT at least once could have introduced selection bias. The study was accepted for review as it was published “online first” in November 2024.

These studies collectively represent a spectrum of approaches, some works (such as those by Acosta-Vargas et al. (2024b)) focus on the technical evaluation of GAI tools applying normative criteria such as WCAG 2.2, while others (such as Liu et al. (2024) and Pierrès et al. (2025)) address real-life user experiences by people with cognitive disabilities or neuro divergences. Also included are developments of specific tools (such as TwIPS (Haroon and Dogar, 2024)) and theoretical reflections on the role of cognitive diversity in the design of intelligent systems (as in Duarte et al. (2024)).

The main objectives of this study were to identify the current characteristics and

the main cognitive accessibility challenges present in GAI interfaces (RQ1), and to map the proposed solutions to improve accessibility for people with cognitive disabilities (RQ2). These results are systematically analyzed through the lens of our conceptual framework, focusing on how design elements impact cognitive load, predictability, and user scaffolding. The findings based on each of the posed research questions are presented below.

3.1. Characteristics and Cognitive Accessibility Challenges in GAI interfaces (RQ1)

The reviewed articles describe an evolving landscape in which GAI interfaces are moving towards a user-intent based interaction model (Duarte et al., 2024). In this paradigm, users express what they want to achieve, and the system executes the steps it considers as appropriate for delivering the result. This model simplifies certain processes but inherently introduces excessive cognitive load (Acosta-Vargas et al., 2024a; Pierrès et al., 2025; Roomkham and Sitbon, 2024). This overload frequently arises from long or dense responses, complex vocabulary, and the rapid presentation of information, factors that significantly hinder comprehension, especially for individuals with ADHD or attention and memory difficulties, as they place a high load on working memory.

The need to formulate precise and effective instructions (a skill commonly referred to as prompt engineering) becomes a core skill in itself that not all users possess (Duarte et al., 2024; Liu et al., 2024). This requirement for complex prompt engineering represents a significant barrier, disproportionately affecting users with cognitive disabilities and creating inequalities in access to the benefits of AI, as it reveals a lack of adequate user scaffolding in the interface design (Duarte et al., 2024; Roomkham and Sitbon, 2024).

A recurrent theme in the reviewed literature is the intrinsic complexity and low transparency of GAI systems (Acosta-Vargas et al., 2024b; Duarte et al., 2024). Even when the interfaces appear simple on the surface, the underlying algorithmic processes are often opaque and unpredictable, limiting users' ability to understand how decisions are made. This lack of behavioral predictability violates the core HCI principle of predictability and contributes to user confusion, mental fatigue, and mistrust (Pierrès et al., 2025). Furthermore, this opacity is coupled with the risk of generating inaccurate results or "hallucinations" which further compromises system predictability and complicates the necessary task of verifying the accuracy of the generated information.

In terms of interaction design, most interfaces are based on text or voice (Duarte et al., 2024; Roomkham and Sitbon, 2024). While conversational design can facilitate use for certain profiles, this limitation highlights the lack of flexibility for users who require alternative communication styles, such as visual, symbolic, or simplified formats (Liu et al., 2024; Haroon and Dogar, 2024). This reliance on text-based modalities excludes nonverbal users and those with literal or visual thinking styles, representing a failure to provide adequate scaffolding and input flexibility for diverse cognitive needs.

Several studies point out that many of these tools have been developed without explicitly considering users with cognitive disabilities, thereby reproducing structural exclusions common in the broader history of digital interface design (Liu et al., 2024; Duarte et al., 2024; Pierrès et al., 2025). Certain emerging features in tools such as ChatGPT have demonstrated potential to support people with dyslexia, ADHD, or ASD by offering summaries, text simplification, organizational assistance, or personalized messages (Pierrès et al., 2025; Haroon and Dogar, 2024). However, these features are

often incidental and not always explicitly integrated into accessible design, meaning they function as incidental supports rather than systematic scaffolding mechanisms (Pierrès et al., 2025).

Regarding compliance with accessibility standards, particularly WCAG 2.2, the reviewed studies report mixed results (Acosta-Vargas et al., 2024a,b). Some interfaces demonstrate strong structural elements and appropriate use of semantic labeling, while others fail to meet basic requirements such as consistent navigation or language specification. These inconsistencies suggest a lack of systematic integration of accessibility principles during the development of many GAI tools, leading to additional usability issues such as unclear instructions, inconsistent navigation, and limited user control over features (e.g., organizing chats or managing time of use), and the challenges posed by the rapid evolution of GAI tools. The challenges posed by the rapid evolution of GAI tools further reduces users' sense of mastery and confidence, reinforcing the lack of predictability inherent in the system's development lifecycle (Duarte et al., 2024; Pierrès et al., 2025; Liu et al., 2024; Acosta-Vargas et al., 2024a,b).

Finally, the review highlights concerns about data privacy (Duarte et al., 2024; Pierrès et al., 2025) and the potential for functional dependency (Pierrès et al., 2025; Haroon and Dogar, 2024). The necessity to collect large volumes of data for AI model training increases privacy violation risks, which is particularly significant for people with disabilities. Over-reliance on GAI for organizing or problem-solving could also have long-term effects on cognitive skill development.

These combined challenges highlight the urgent need for inclusive design practices that address not only surface-level accessibility, but also deeper cognitive and learning-related barriers, specifically focusing on the management of cognitive load to reduce overload and optimizing mental operations. If the system is predictable, the user does not become overwhelmed and with the appropriate scaffolding, it guides the user to perform tasks with less cognitive overload.

3.2. Solutions proposed to improve the accessibility of textual GAI interfaces for individuals with cognitive disabilities (RQ2)

The solutions identified in the reviewed articles can be grouped into six major lines of action aimed at enhancing the cognitive accessibility of GAI interfaces.

The first line emphasizes the importance of inclusive design from the earliest stages of design and development (Acosta-Vargas et al., 2024a,b; Duarte et al., 2024). This involves integrating principles of accessibility, equity, and inclusion into interaction design, and ensuring the active participation of people with disabilities in both design and evaluation processes (Pierrès et al., 2025; Acosta-Vargas et al., 2024a; Haroon and Dogar, 2024; Liu et al., 2024).

The second line defends the development of multimodal, intuitive, and explainable interfaces. These systems should integrate diverse input and output modalities (voice, text, images, gestures, simplified visualization) (Roomkham and Sitbon, 2024; Pierrès et al., 2025) and promote iterative interaction, allowing users to refine outputs through dialogue (Liu et al., 2024). Support tools like tutorials (Liu et al., 2024; Acosta-Vargas et al., 2024a), clear confirmations, and content summaries are recommended (Acosta-Vargas et al., 2024a). These actions will provide fundamental user scaffolding and reduce the cognitive load of users. Furthermore, promoting transparency and explainability of AI models (Duarte et al., 2024; Acosta-Vargas et al., 2024a; Haroon and Dogar, 2024) is key to restoring predictability and increasing user understanding and autonomy.

The third proposal focuses on personalization and adaptation, with systems that can adjust the complexity, tone, and presentation of content to match users’ cognitive and communicative profiles (Pierrès et al., 2025; Roomkham and Sitbon, 2024; Acosta-Vargas et al., 2024a; Haroon and Dogar, 2024). This includes offering targeted assistance (e.g., spelling support or visual cues), and dynamically tailoring interactions to users’ needs and preferences. This is directly aimed at managing cognitive load and represents an individualized form of scaffolding adapted to users’ cognitive profiles.

The fourth line aims to reduce technological and cognitive barriers. It highlights the need to simplify interaction flows, remove unnecessary distractions, streamline access and registration processes, and develop solutions that can operate without requiring continuous connectivity or advanced hardware. The integration of built-in accessibility features, such as accessible previews or automated accessibility checks, is also emphasized (Liu et al., 2024; Duarte et al., 2024). These measures are essential for effective cognitive load management and establishing consistent, predictable user experiences.

A fifth direction promotes the creation of collaborative and context-aware platforms. These systems would support peer interaction and co-creation while incorporating mechanisms to understand user intent and context, using cues such as conversation topics or nonverbal signals, thereby enabling more responsive and meaningful user experiences (Roomkham and Sitbon, 2024; Duarte et al., 2024; Acosta-Vargas et al., 2024a).

Finally, the literature highlights the importance of establishing a comprehensive ethical, educational, and regulatory framework. This includes developing equity and transparency metrics, offering accessibility training for developers, encouraging participatory dialogue with marginalized communities, ensuring compliance with standards such as WCAG, and promoting research agendas that prioritize social needs over purely commercial interests (Acosta-Vargas et al., 2024a,b; Duarte et al., 2024; Pierrès et al., 2025).

Together, these proposals outline a roadmap toward more inclusive and cognitively accessible GAI systems, an area where practical implementation remains limited, but where the reviewed studies suggest both significant challenges and opportunities for progress.

4. Discussion

The core finding of this review is that cognitive accessibility remains a critical, yet largely unmet, challenge in textual GAI interfaces. The current user-intent based interaction model, characterized by its reliance on prompt engineering, low explainability, and text-centric modalities, violates core principles of cognitive HCI: it places an excessive cognitive load on users, severely compromises predictability, and demonstrates a systemic lack of user scaffolding. These three inherent characteristics may undermine usability for individuals with cognitive disabilities. While GAI has substantial potential to support these users through features like simplification and organizational assistance, these beneficial characteristics are often incidental, rather than systematically integrated into inclusive design practices.

The most critical gap lies between the identified challenges and the proposed solutions. While the literature effectively maps the theoretical violation of cognitive load management, predictability, and scaffolding, the proposed remedies – including multimodal and explainable interfaces, adaptive content, and the advocacy for ethical frameworks and accessibility-aware policies (Duarte et al., 2024) – remain largely fragmented and conceptual. These solutions are explicitly designed to restore these principles. For instance, multimodal modalities improve comprehension (Liu et al., 2024; Roomkham

and Sitbon, 2024) by providing crucial scaffolding and reducing verbal cognitive load. Nonetheless, these solutions are often implemented only at a prototype level or in controlled research settings, with little evidence of real-world deployment or large-scale validation. This absence of deployment is crucial, as the use of GAI currently requires high critical thinking and discernment, which represents an extra cognitive burden for many users. But the truth is that cognitive load management, as the general framework for optimizing mental processing, may provide less overload and greater clarity for users with cognitive disabilities, predictability can help to reduce anxiety and decrease the need for inference, and user scaffolding can provide greater autonomy and support guided learning. Together, the three characteristics may create accessible and cognitively sustainable environments when they are conveniently integrated.

4.1. Strengths and Limitations

This review presents several notable strengths, particularly given the emerging nature of the field. One of the primary strengths lies in its timely and specific focus on a highly relevant but underexplored intersection: that of cognitive accessibility and GAI interface design. As one of the first systematic efforts to explore this rapidly evolving domain, this study provides a crucial foundational synthesis of the existing literature.

By employing a rigorous methodology based on the PRISMA framework and drawing from four major academic databases, the review offers a transparent and representative overview of current knowledge published between 2020 and 2024. Significantly, all studies that met the inclusion criteria were published in 2024. This absence of relevant publications in earlier years can be explained by the timeline of public engagement with GAI technologies. Although foundational models like GPT-3 were introduced in 2020, their implications for interface design and accessibility only became tangible following the widespread adoption of tools like ChatGPT from late 2022 onward. In the initial years, GAI was primarily explored in technical or conceptual terms, without concrete applications that intersected with cognitive accessibility. As a result, academic inquiry into this specific domain only began to emerge once GAI tools became part of everyday digital experiences.

In addition, the use of clearly defined and rigorous inclusion criteria, centered specifically on text-based GAI interfaces and cognitive accessibility, ensures the high relevance and sharp focus of the selected studies. This precise scope enables an in-depth analysis of specific challenges and emerging solutions, yielding actionable insights for both researchers and practitioners. This systematic approach effectively maps the early landscape, identifying critical research gaps and providing a clear roadmap for future, more robust investigations into GAI for cognitive diversity.

Nevertheless, several limitations constrain the generalizability of the findings. First, this systematic review is based on a limited number of studies ($n = 7$), which reflects the early stage of research in this area.

Second, the results are limited by the quality and methodological approaches of the included studies. While all of them contribute to understanding the relationship between GAI and cognitive accessibility, they vary widely in scope and rigor. The studies employ highly disparate approaches, such as automated audits (Acosta-Vargas et al., 2024a,b), qualitative interviews (Liu et al., 2024; Pierrès et al., 2025), prototype evaluations (Roomkham and Sitbon, 2024; Haroon and Dogar, 2024), and conceptual analyses (Duarte et al., 2024).

Moreover, those studies involving real users often rely on small or unrepresentative

samples, frequently focusing on a single group (Liu et al., 2024; Roomkham and Sitbon, 2024; Haroon and Dogar, 2024; Pierrès et al., 2025). Many evaluated systems are in early development stages, including prototypes not tested in real-world environments or manually operated by researchers.

Additionally, the conceptualization of cognitive accessibility is often limited. Some studies focus on specific conditions, such as autism, dyslexia, or ADHD, without addressing cognitive diversity more broadly. Very few works include participants with moderate or severe intellectual disabilities, which leaves out a particularly vulnerable group in terms of digital accessibility (Liu et al., 2024; Roomkham and Sitbon, 2024; Pierrès et al., 2025).

Taken together, these limitations underscore that research on cognitive accessibility in GAI interfaces is still in its early stages. We need to move toward more integrative studies, with rigorous methodological approaches, greater participation of users with diverse profiles, and deeper attention to cognitive barriers, both from the design and evaluation of the actual use of these technologies.

4.2. Implications for Practice and Research

The results of this systematic review show that cognitive accessibility in GAI interfaces is an emerging field, with multiple knowledge gaps and plenty of opportunities for future research. Based on the analysis of the reviewed studies, the following priority lines are proposed.

First, there is an urgent need to conduct more empirical studies, particularly those focused on end-users with different types of cognitive or intellectual disabilities. Currently, most studies include small samples, very specific profiles, or do not directly incorporate people with disabilities in the evaluation processes. Conducting user studies, usability tests, and longitudinal analyses in real-life contexts is essential to understand the specific barriers faced by these groups and adapt technological solutions to their needs.

Also, the development and validation of specific evaluation methods for cognitive accessibility in GAI tools must be a priority. While standards such as WCAG exist, their general and technical approach does not always allow for adequate assessment of the key cognitive factors identified in this review: cognitive load, predictability, and scaffolding (Acosta-Vargas et al., 2024a,b). In this sense, the guidelines developed by the W3C’s COGA task force (COGA Working Group, 2020) represent a significant advance, proposing specific recommendations for improving the cognitive accessibility of web content and designs. However, these recommendations still need to be transformed into systematic evaluation tools, with validated objective/subjective metrics and procedures that include the direct experience of people with cognitive disabilities in their design and application. This is especially critical in the context of GAI interfaces, whose complexity and unpredictability require more specific and sensitive evaluation approaches.

Another key line of research is the proposal and validation of specific design guidelines for cognitive accessibility in GAI interfaces. Unlike other more established technologies, GAI interfaces have specific characteristics, such as natural language-based interaction, that require new guidelines adapted to the emerging paradigm. These guidelines should be structured around HCI principles to ensure actionable design decisions. They should specifically address how to minimize cognitive load through multimodal output and predictions to reduce choice overload, how to maximize predictability through clearer model explainability, easy navigation and uniform terminology and how to embed scaffolding mechanisms like prompt templates or contextual help (Liu et al., 2024; Roomkham and

Sitbon, 2024; Haroon and Dogar, 2024).

Furthermore, future research could expand to include cognitive accessibility in other types of GAI beyond text. Tools that generate images, code, sound, or multimodal experiences also pose significant accessibility challenges that have not yet been sufficiently explored (Liu et al., 2024; Roomkham and Sitbon, 2024). Understanding how people with cognitive disabilities interact with these types of systems will allow us to extend the benefits of GAI to a wider range of users and contexts.

Finally, the importance of exploring ethical, regulatory, and educational dimensions related to cognitive accessibility in GAI is also highlighted. This includes developing ethical frameworks that protect user autonomy, designing inclusive institutional policies, providing specific training for designers and developers in cognitive accessibility, and creating regulations that guarantee inclusion from design to implementation (Duarte et al., 2024).

Taken together, these future directions point to the need to consolidate an interdisciplinary field of research that articulates knowledge from engineering, user-centered design, cognitive science, accessibility, and Artificial Intelligence. Only in this way it will be possible to ensure that generative technologies are truly inclusive and respond to the cognitive diversity of society.

5. Conclusions

This systematic literature review provides a structured and critical analysis of the current state of cognitive accessibility in GAI interfaces, specifically focusing on text-based natural language tools. The review addressed its primary goal by identifying the current design landscape and associated cognitive challenges (RQ1), and synthesizing the proposed solutions (RQ2).

The findings reveal a significant gap: current GAI interface designs predominantly are focused on neurotypical users, exhibiting low transparency and a language-based interaction model that fails to adapt to diverse cognitive styles. The analysis highlights that the current characteristics and main accessibility challenges (RQ1) are intertwined: the design features – such as reliance on complex prompt engineering and the text-centric modality – directly cause barriers, including excessive cognitive load, lack of predictability, and poor system explainability.

In response to the second research question (RQ2), the literature offers promising, though incipient, lines for improvement. These proposed solutions point toward inclusive approaches that prioritize adaptability, transparency, and user participation in the design process, primarily through multimodal interfaces and personalized experiences adapted to the user’s cognitive profile. However, these strategies often remain conceptual or at the prototype stage, underscoring a critical research-to-practice gap.

The main contribution of this article is to offer a critical and updated synthesis of existing knowledge at the intersection of Generative AI and cognitive accessibility, a field that is still underexplored but of great social relevance. We move beyond simple literature mapping by systematically analyzing the identified challenges and proposed solutions through the lens of core HCI principles (cognitive load, predictability, and user scaffolding). This analytical focus provides a clear conceptual foundation for future research and contributes actionable insights for the development of specific standards and truly inclusive GAI technologies, by identifying both theoretical violations and practical gaps.

Addressing cognitive accessibility should not be considered a secondary or comple-

mentary aspect, but rather a fundamental condition for GAI to fulfill its promise of innovation with social justice. Designing for cognitive diversity is key to ensuring that these transformative technologies benefit all people, leaving no one behind.

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Disclosure statement

The authors report there are no competing interests to declare.

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